1. **Problem 1 (7-1)**

   1st condition

$$P_L = 20 \text{ mm Hg} = 2666 \text{ Pa}$$
$$P_H = 1 \text{ atm} = 101325 \text{ Pa}$$
$$y_0 = 0.209$$
$$y_j = 1 \text{ ppm} = 1 \cdot 10^{-6}$$
$$y_j = y_0 \left(\frac{P_L}{P_H}\right)^j$$
$$\ln\left(\frac{y_j}{y_0}\right) = \ln\left(\left(\frac{P_L}{P_H}\right)^j\right)$$
$$j = \frac{\ln\left(\frac{y_j}{y_0}\right)}{\ln\left(\frac{P_L}{P_H}\right)}$$
$$j = \frac{\ln\left(\frac{1\cdot 10^{-6}}{0.209}\right)}{\ln\left(\frac{2666}{101325}\right)}$$
$$\boxed{j = 3.37}$$
$$\Delta n = j\left(P_H - P_L\right)\frac{V}{RT}$$
$$\Delta n = 4 \cdot (101325 - 2666) \cdot \frac{3.78}{8.314 \cdot 297}$$
$$\Delta n = 603.8 \text{ mol}$$
$$m = \frac{603.8 \cdot 28}{1000}$$
$$\boxed{m = 16.9 \text{ kg}}$$

   2nd condition

$$P_L = 30 \text{ mm Hg} = 4000 \text{ Pa}$$

$$P_H = 1 \text{ atm} = 101325 \text{ Pa}$$

$$y_0 = 0.209$$

$$j = \frac{\ln\left(\frac{1\cdot 10^{-6}}{0.209}\right)}{\ln\left(\frac{4000}{101325}\right)}$$

$$\boxed{j = 3.79}$$

$$\Delta n = j\left(P_H - P_L\right)\frac{V}{RT}$$

$$\Delta n = 4 \cdot (101325 - 4000) \cdot \frac{37}{8.314 \cdot 297}$$

$$\Delta n = 5830.4 \text{ mol}$$

$$m = \frac{5830.4 \cdot 28}{1000}$$

$$\boxed{m = 163.3 \text{ kg}}$$

2. **Problem 2 (7-2)**

   1st condition

$$
\begin{aligned}
P_H &= 4136 \text{ mm Hg} = 551419 \text{ Pa} \\
P_L &= 1 \text{ atm} = 101325 \text{ Pa} \\
y_0 &= 0.209 \\
j &= \frac{\ln\left(\frac{1 \cdot 10^{-6}}{0.209}\right)}{\ln\left(\frac{101325}{551419}\right)} \\
&\boxed{j = 7.23} \\
\Delta n &= j\left(P_H - P_L\right)\frac{V}{RT} \\
\Delta n &= 8 \cdot (551419 - 101325) \cdot \frac{3.78}{8.314 \cdot 297} \\
\Delta n &= 5509.3 \text{ mol} \\
m &= \frac{5509.3 \cdot 28}{1000} \\
&\boxed{m = 154.3 \text{ kg}}
\end{aligned}
$$

   2nd condition

$$
\begin{aligned}
P_H &= 3000 \text{ mm Hg} = 399966 \text{ Pa} \\
P_L &= 1 \text{ atm} = 101325 \text{ Pa} \\
y_0 &= 0.209 \\
j &= \frac{\ln\left(\frac{1 \cdot 10^{-6}}{0.209}\right)}{\ln\left(\frac{101325}{399966}\right)} \\
&\boxed{j = 8.92} \\
\Delta n &= j\left(P_H - P_L\right)\frac{V}{RT} \\
\Delta n &= 9 \cdot (399966 - 101325) \cdot \frac{37}{8.314 \cdot 297} \\
\Delta n &= 40253 \text{ mol} \\
m &= \frac{40253 \cdot 28}{1000} \\
&\boxed{m = 1127.7 \text{ kg}}
\end{aligned}
$$

3. **Problem 3 (7-7)**

a.

$$t = \frac{1800 \cdot 1.89}{5} = 680.4 \text{ s}$$
$$I_S = 5 \cdot 10^{-7}$$
$$Q = I_S t$$
$$Q = 5 \cdot 10^{-7} \cdot 680.4$$
$$\boxed{Q = 3.40 \cdot 10^{-4} \text{ C}}$$
$$J = \frac{Q^2}{2C}$$
$$J = \frac{(3.40 \cdot 10^{-4})^2}{2 \cdot 8.54 \cdot 10^{-11}}$$
$$\boxed{J = 6.77 \cdot 10^{2} \text{ J}}$$
$$\boxed{J = 6.77 \cdot 10^{5} \text{ mJ}}$$

Above 10 mJ MIE for dust.

b.

$$I_S = 5 \cdot 10^{-5}$$
$$Q = I_S t$$
$$Q = 5 \cdot 10^{-5} \cdot 680.4$$
$$\boxed{Q = 3.40 \cdot 10^{-2} \text{ C}}$$
$$J = \frac{Q^2}{2C}$$
$$J = \frac{(3.40 \cdot 10^{-2})^2}{2 \cdot 8.54 \cdot 10^{-11}}$$
$$\boxed{J = 6.77 \cdot 10^{6} \text{ J}}$$
$$\boxed{J = 6.77 \cdot 10^{9} \text{ mJ}}$$

Above 10 mJ MIE for dust.